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Oefeningenreeks 5G nr. 3.1

$$f(x) = e^{-x} \text{ op } [0, +\infty[$$

$$\begin{aligned} V &= \pi \lim_{t \rightarrow +\infty} \int_0^t e^{-2x} dx \\ &= \pi \lim_{t \rightarrow +\infty} \left[\frac{-1}{2} e^{-2x} \right]_0^t \\ &= \pi \lim_{t \rightarrow +\infty} \left(\frac{-e^{-2x}}{2} + \frac{1}{2} \right) = \frac{\pi}{2} \end{aligned}$$

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References